Browser Forensic Artifact Analysis Report

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Tools Used: DB Browser for SQLite, Hindsight, Google Chrome
Platform: Windows 10

1. Introduction

Browser forensic analysis is an important component of Digital Forensics and Incident Response (DFIR), enabling investigators to reconstruct user activities through digital artifact examinations.

The objective of this project was to analyze Google Chrome browser artifacts to extract and evaluate browsing history, downloads, search queries, and related user behavior data.

This project demonstrates practical knowledge of forensic evidence handling, database artifact parsing, and activity reconstruction techniques.

This analysis also considered the relevance of artifacts to potential security investigations, highlighting patterns that could indicate policy violations, data exfiltration, or abnormal user behavior.
Observations were documented with the intention of supporting both forensic evidence integrity and actionable security insights.

2. Objective

The primary objectives of this project were:

- To acquire artifact evidence safely.
- To perform manual and automated artifact extraction.
- To analyze browsing behavior patterns.
- To reconstruct a timeline of user activity.
- To validate automated forensic tool results.

3. Tools and Environment

3.1 Software Tools

- DB Browser for SQLite – Manual database artifact inspection.
- Hindsight – Automated Chrome artifact forensic analysis tool.
- Google Chrome – Source of browser data.

3.2 Operating Environment

- Host Operating System: Windows 10
- Target Data Source: Chrome user profile directory

4. Evidence Acquisition Process

Prior to analysis, Google Chrome was completely closed to prevent database locking.

The browser history database file was in the Chrome user profile directory:

C:\Users\[Username]\AppData\Local\Google\Chrome\User Data\Default

The History database file was copied to a working directory to ensure preservation of original evidence.

This approach aligns with standard digital forensic practice of preserving original evidence integrity by performing analysis on copied datasets.

5. Artifact Analysis Methodology

5.1 Manual Database Analysis

The Chrome History SQLite database was examined using DB Browser for SQLite.

The investigation focused on the urls table, which stores browsing activity metadata.

The following fields were analyzed:

- url – Visited website address
- title – Webpage title associated with the URL
- visit_count – Number of times a website was accessed
- last_visit_time – Chrome internal timestamp format measured in microseconds since 1601-01-01

Manual inspection was performed to support verification of automated parsing results.

5.2 Automated Artifact Analysis Using Hindsight

Automated forensic parsing was performed using Hindsight tool.

The tool was executed locally, and the web interface was accessed at:
http://localhost:8080

The Chrome Default profile directory was provided as input.

Hindsight processed browser database files and generated structured forensic outputs.

Extracted artifact categories included:

- Browsing history records
- Download activity logs
- Search query records
- Bookmark information
- Autofill form entries
- Extension-related metadata

6. Behavioral Observation

The artifact analysis suggests browsing activity across multiple domain categories, including educational, professional, and entertainment platforms.

The presence of a large number of search queries suggests active information retrieval behavior. However, search volume alone does not indicate abnormal or malicious activity.

Repeated access to specific websites may indicate user preference patterns or habitual browsing behavior.

The analysis was primarily focused on artifact extraction from browser history data. Detailed behavioral anomaly detection, threat correlation, or baseline user profiling was outside the scope of this study.

7. Findings

7.1 Artifact Extraction Summary

Artifact Type	Count
URL records	18,182
Download records	650
Google search queries	6,790
Bookmarks	182
Autofill entries	578

7.2 Behavioral Interpretation

- Browsing activity shows repeated access to selected websites.
- Search query volume indicates active information exploration.
- Download records suggest retrieval of external documents or media files.
- Bookmark usage reflects user preference for quick resource access.

8. Validation of Analysis

- Results obtained from automated parsing were cross-validated with manual database inspection.
- Consistency between tool output and SQLite record examination improves reliability of forensic findings.

9. Limitations

- Analysis was restricted to Chrome Default profile artifacts.
- Network traffic, memory artifacts, and encrypted data were not examined.
- The investigation was carried out in a controlled laboratory environment.

10. Conclusion

This project demonstrated practical browser forensic artifact analysis using both manual SQLite database inspection and automated parsing techniques.

The investigation successfully extracted thousands of browsing history records, search queries, and downloaded entries.

The study primarily focused on artifact extraction and behavioural observation rather than advanced timeline reconstruction or anomaly detection.

Limitations of this work include the exclusion of memory forensics, network traffic analysis, and encrypted data examination.

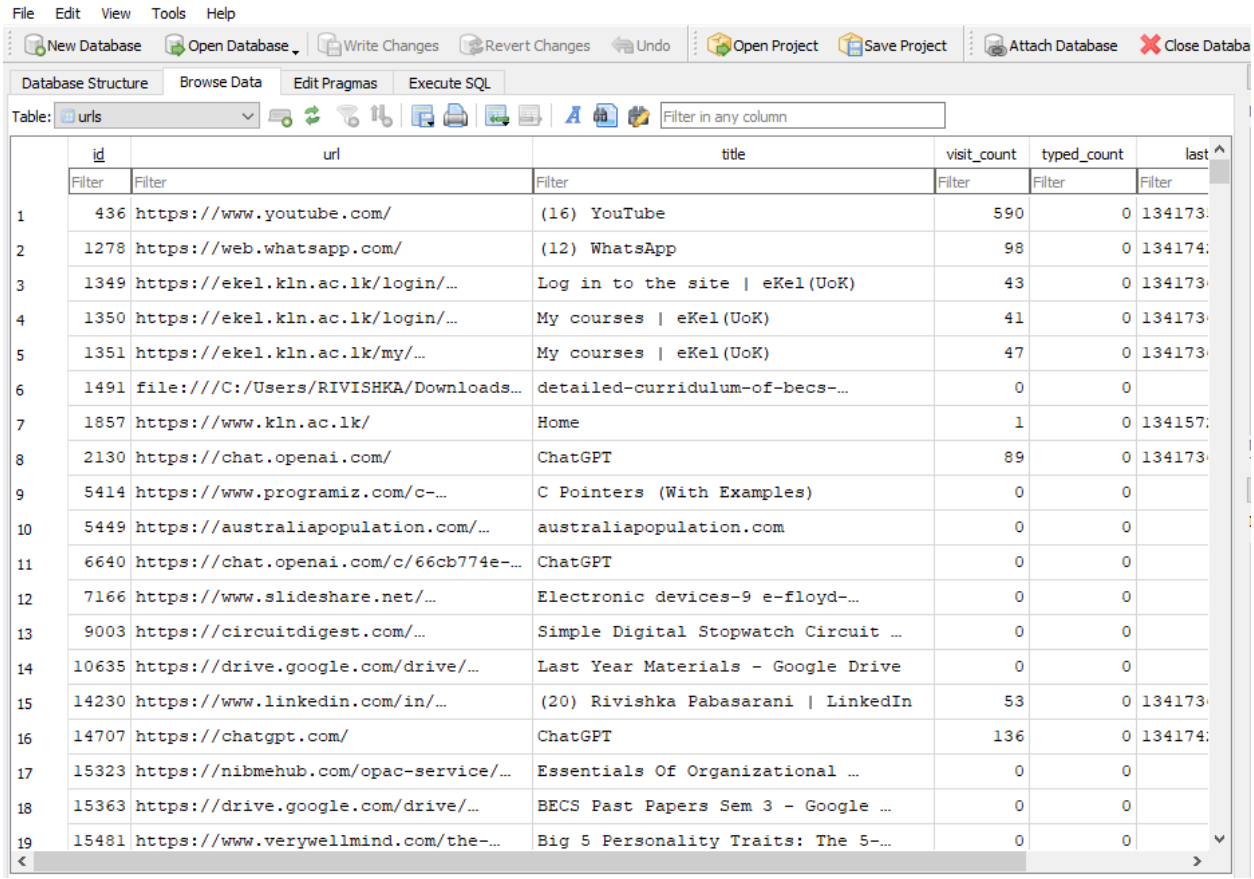
Future work may include multi-browser artifact correlation, behavioural baseline modelling, and advanced threat intelligence integration.

11. Skills Demonstrated

- Digital Forensics Investigation Process
- SQLite Database Artifact Analysis
- Browser History Reconstruction
- DFIR Tool Usage
- Behavioral Evidence Interpretation
- Technical Documentation

Appendices

Appendix A: DB Browser – urls table view



File Edit View Tools Help

New Database | Open Database | Write Changes | Revert Changes | Undo | Open Project | Save Project | Attach Database | Close Databa

Database Structure | Browse Data | Edit Pragmas | Execute SQL

Table: urls | Filter in any column

	id	url	title	visit_count	typed_count	last
	Filter	Filter	Filter	Filter	Filter	Filter
1	436	https://www.youtube.com/	(16) YouTube	590	0	134173
2	1278	https://web.whatsapp.com/	(12) WhatsApp	98	0	134174
3	1349	https://ekel.kln.ac.lk/login/…	Log in to the site | eKel(UoK)	43	0	134173
4	1350	https://ekel.kln.ac.lk/login/…	My courses | eKel(UoK)	41	0	134173
5	1351	https://ekel.kln.ac.lk/my/…	My courses | eKel(UoK)	47	0	134173
6	1491	file:///C:/Users/RIVISHKA/Downloads…	detailed-curridulum-of-becs-…	0	0	
7	1857	https://www.kln.ac.lk/	Home	1	0	134157
8	2130	https://chat.openai.com/	ChatGPT	89	0	134173
9	5414	https://www.programiz.com/c-…	C Pointers (With Examples)	0	0	
10	5449	https://australiapopulation.com/…	australiapopulation.com	0	0	
11	6640	https://chat.openai.com/c/66cb774e-…	ChatGPT	0	0	
12	7166	https://www.slideshare.net/…	Electronic devices-9 e-floyd-…	0	0	
13	9003	https://circuitdigest.com/…	Simple Digital Stopwatch Circuit …	0	0	
14	10635	https://drive.google.com/drive/…	Last Year Materials - Google Drive	0	0	
15	14230	https://www.linkedin.com/in/…	(20) Rivishka Pabasarani | LinkedIn	53	0	134173
16	14707	https://chatgpt.com/	ChatGPT	136	0	134174
17	15323	https://nibmehub.com/opac-service/…	Essentials Of Organizational …	0	0	
18	15363	https://drive.google.com/drive/…	BECS Past Papers Sem 3 - Google …	0	0	
19	15481	https://www.verywellmind.com/the-…	Big 5 Personality Traits: The 5-…	0	0	

Appendix B: Hindsight – main results summary

Summary

Input Path: C:\Users\[redacted]\AppData\Local\Google\Chrome\User Data\Default

Input Type: Chrome

Profile Paths:

- C:\Users\[redacted]\AppData\Local\Google\Chrome\User Data\Default

Plugin Results

Chrome Extension Names [v20240428]:	- 35 extension URLs parsed -
Generic Timestamp Decoder [v20240428]:	- 0 timestamps parsed -
Google Analytics Cookie Parser [v20170130]:	- 0 cookies parsed -
Google Searches [v20160912]:	- 6790 searches parsed -
Load Balancer Cookie Decoder [v20200213]:	- 0 cookies parsed -
Quantcast Cookie Parser [v20160907]:	- 0 cookies parsed -
Query String Parser [v20170225]:	- 6336 query strings parsed -
Time Discrepancy Finder [v20170129]:	- 0 differences parsed -

Appendix C: Hindsight – plugins and extensions

```
Profile
Path: C:\Users\[redacted]\AppData\Local\Google\Chrome\User Data\Default
Detected Browser: Chrome v142-144

User Activity                                Count
                          URL records    [   18182 ]
                     Download records    [     650 ]
                     Autofill records    [     578 ]
                  Login Data records     [      82 ]
         Login Data (Account) records    [       0 ]
                     Bookmark records    [     182 ]

Website Storage                              Count
                       Cookie records    [  Failed ]
                        Cache records    [  Failed ]
                    GPU Cache records    [  Failed ]
                Local Storage records    [   39825 ]
              Session Storage records    [     388 ]
                    IndexedDB records    [  135735 ]
                    File System items    [      24 ]

Browser Extensions                           Count
                 Installed Extensions    [       3 ]
            Extension Cookie records     [       0 ]
             Extension Rules records     [       0 ]
           Extension Scripts records     [   16840 ]
             Extension State records     [   35962 ]
      Local Extension Settings records   [    5277 ]
            Sync App Settings records    [       0 ]
       Sync Extension Settings records   [       0 ]

Configuration & Supporting Data              Count
                     Preference items    [    2125 ]
          Site Characteristics records   [   12070 ]
                    Sync Data records    [    3804 ]
                         HSTS records    [    2079 ]
                  DIPS Popup records     [      18 ]
                         DIPS records    [     256 ]

Running Plugins
```

```
Running Plugins

                 Chrome Extension Names (v20240428):   - 35 extension URLs parsed -
             Generic Timestamp Decoder (v20240428):     - 0 timestamps parsed -
       Google Analytics Cookie Parser (v20170130):       - 0 cookies parsed -
                 Google Searches (v20160912):     - 6790 searches parsed -
           Load Balancer Cookie Decoder (v20200213):       - 0 cookies parsed -
                Quantcast Cookie Parser (v20160907):       - 0 cookies parsed -
                    Query String Parser (v20170225):  - 6336 query strings parsed -
                Time Discrepancy Finder (v20170129):     - 0 differences parsed -
127.0.0.1 - - [08/Mar/2026 11:37:59] "POST /run HTTP/1.1" 303 0
127.0.0.1 - - [08/Mar/2026 11:38:00] "GET /results HTTP/1.1" 200 13518
127.0.0.1 - - [08/Mar/2026 11:38:00] "GET /static/styles.css HTTP/1.1" 304 0
127.0.0.1 - - [08/Mar/2026 11:38:00] "GET /static/github.png HTTP/1.1" 304 0
127.0.0.1 - - [08/Mar/2026 11:38:00] "GET /static/mastodon.svg HTTP/1.1" 304 0
127.0.0.1 - - [08/Mar/2026 11:38:00] "GET /static/h.png HTTP/1.1" 304 0
127.0.0.1 - - [08/Mar/2026 11:38:01] "GET /static/icons/manifest.json HTTP/1.1" 304 0
```